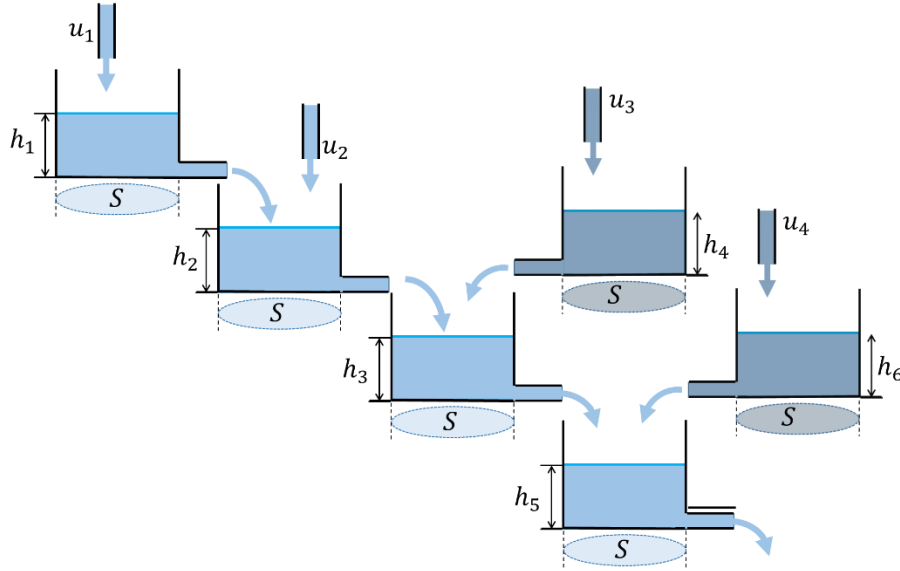


Six-tanks system 01

Consider the system illustrated in the following figure, consisting of a cascade interconnection of six tanks.



The linearized centralized model is

$$\begin{cases} S\dot{h}_1 = -kh_1 + u_1 \\ S\dot{h}_2 = kh_1 - kh_2 + u_2 \\ S\dot{h}_3 = kh_2 + kh_4 - kh_3 \\ S\dot{h}_4 = -kh_4 + u_3 \\ S\dot{h}_5 = kh_3 + kh_6 - kh_5 \\ S\dot{h}_6 = -kh_6 + u_4 \end{cases}$$

where $S = 1 \text{ m}^2$ and $k = 1 \text{ m}^2/\text{s}$. Note that, since the model is linearized around a nominal condition, all the variables of the model above should be regarded as differences with respect to nominal values. We assume that all levels are measurable. Defining $x = [h_1, \dots, h_6]^T$ and $u = [u_1, u_2, u_3, u_4]^T$, the system's dynamics is described by the model

$$\dot{x} = Ax + Bu$$

where the matrices are defined in the corresponding MATLAB file.

Problem:

1. Decompose the state and input vectors into subvectors, consistently with the physical description of the system. Obtain the corresponding decomposed model.
2. Generate the system matrices (both continuous-time and discrete-time, the latter with a sampling time selected compatibly with the continuous-time dynamics). Perform the following analysis:
 - a. Compute the eigenvalues and the spectral abscissa of the (continuous-time) system. Is it open-loop asymptotically stable?
 - b. Compute the eigenvalues and the spectral radius of the (discrete-time) system. Is it open-loop asymptotically stable?
3. For different state-feedback control structures (i.e., centralized, decentralized, and different distributed schemes) perform the following actions
 - a. Compute the continuous-time fixed modes
 - b. Compute the discrete-time fixed modes
 - c. Compute, if possible, the CONTINUOUS-TIME control gains using LMIs to achieve the desired performances. Apply, for better comparison, different criteria for computing the control laws.
 - d. Compute, if possible, the DISCRETE-TIME control gains using LMIs to achieve the desired performances. Apply, for better comparison, different criteria for computing the control laws.
 - e. Analyze the properties of the so-obtained closed-loop systems (e.g., stability, eigenvalues) and compute the closed-loop system trajectories (generated both in continuous-time and in discrete-time) of the water levels starting from a common random initial condition.